

Why Scenario C (multi-prompt item means) captures $\sigma_{\alpha\rho}^2$ but not σ_{ρ}^2

Notes for Figure 1 simulation (04g_sim_underestimation.R)

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The core insight: `sd(item means)` measures variability *across items within a simulation*. Only components that vary across items at fixed $(v_{\text{set}}, h_c, m_c)$ contribute.

Derivation

The item mean under scenario C (multi-prompt item means, $V = 3$, $R = 5$, fixed h_c, m_c) is:

$$\begin{aligned} \bar{y}_{i,\cdot} = & \mu + \alpha_i + \underbrace{\bar{\rho}}_{\text{avg over } v} + \tau[h_c] + \lambda[m_c] \\ & + \underbrace{\overline{(\alpha\rho)_i}}_{\text{avg over } v \text{ for item } i} + (\alpha\tau)_{i,h_c} + (\alpha\lambda)_{i,m_c} \\ & + \underbrace{\overline{(\rho\tau)_{\cdot,h_c}}}_{\text{avg over } v} + \underbrace{\overline{(\rho\lambda)_{\cdot,m_c}}}_{\text{avg over } v} + \bar{\epsilon}_i. \end{aligned}$$

Now sort each term by whether it varies across items within one simulation.

Varies across items (contributes to `sd(item means)`).

- $\alpha_i \longrightarrow \sigma_{\alpha}^2$
- $\overline{(\alpha\rho)_i} \longrightarrow \sigma_{\alpha\rho}^2/V$ (*this is the key one*)
- $(\alpha\tau)_{i,h_c} \longrightarrow \sigma_{\alpha\tau}^2$
- $(\alpha\lambda)_{i,m_c} \longrightarrow \sigma_{\alpha\lambda}^2$
- $\bar{\epsilon}_i \longrightarrow \sigma_{\epsilon}^2/(VR)$

Constant across items in one simulation (invisible to `sd(item means)`).

- $\bar{\rho}$ — same for every item in the simulation; just shifts all item means by the same amount
- $\overline{(\rho\tau)_{\cdot,h_c}}$ — same
- $\overline{(\rho\lambda)_{\cdot,m_c}}$ — same
- $\tau[h_c], \lambda[m_c]$ — deterministic constants

The key distinction

Any term with an i **subscript** that remains after averaging over v varies across items $\rightarrow$ captured by `sd(item means)`.

Any term **without an i subscript** becomes a constant offset after averaging $\rightarrow$ invisible.

- $(\alpha\rho)_{i,v}$ has both i and v . Averaging over v gives $\overline{(\alpha\rho)_i}$, which *still has i* and varies across items. Its contribution shrinks from $\sigma_{\alpha\rho}^2$ to $\sigma_{\alpha\rho}^2/V$ (because we averaged V iid draws per item) but stays nonzero. **Captured.**
- ρ_v has only v . Averaging over v gives $\bar{\rho}$, a scalar. Every item in the simulation gets the same $\bar{\rho}$. Across items: zero variance. **Blind.**

Where σ_ρ^2 actually lives

In the *bias*, not the SE. Across simulations:

$$\bar{\rho} \sim \mathcal{N}(0, \sigma_\rho^2/V).$$

This shifts the entire set of item means by a per-simulation constant. The CI around `mean(item means)` doesn't include this shift. So each simulation's CI is off-center by $\bar{\rho}$, and coverage of the grand mean drops.

C gets the *smaller* blind spot σ_ρ^2/V (because averaging shrinks the shift variance) but still misses it entirely—because the estimator sees variance *across items within a simulation*, not variance *across simulations*.

Implications for Panel (a)

C should show:

- α, γ, ϵ : captured
- $\alpha \times \rho$: captured (this is what's new vs B)
- $\rho, \rho \times \tau, \rho \times \lambda$: still blind (captured only as cross-simulation bias, not in-sample SE)
- $\alpha \times \tau, \alpha \times \lambda$: captured

Scenario B has $V = 1$, so the item mean simplifies to:

$$\bar{y}_{i,\cdot} = \mu + \alpha_i + \rho_{v_c} + \tau[h_c] + \lambda[m_c] + (\alpha\rho)_{i,v_c} + (\alpha\tau)_{i,h_c} + (\rho\tau)_{v_c,h_c} + (\alpha\lambda)_{i,m_c} + (\rho\lambda)_{v_c,m_c} + \bar{\epsilon}_i.$$

Same structure: anything with i subscript contributes to `sd(item means)`. So B captures $\alpha, (\alpha\rho)$ at the chosen v , $(\alpha\tau)$ at the chosen h , $(\alpha\lambda)$ at the chosen m , and ϵ .

B and C both capture $\alpha, \alpha \times \rho, \alpha \times \tau, \alpha \times \lambda$ in the numerical SE—the only difference is that C's contribution for $\alpha \times \rho$ is $\sigma_{\alpha\rho}^2/V$ (averaged over the chosen variants) instead of $\sigma_{\alpha\rho}^2$ (at the single v). But that difference is small for the grand-mean estimand because the true contribution of $\alpha \times \rho$ to $\text{Var}(\hat{\theta})$ is $\sigma_{\alpha\rho}^2/(NV)$ regardless.

This means the earlier Panel (a) classification was wrong: $\alpha \times \tau$ and $\alpha \times \lambda$ were marked “blind” for A/B but should be “captured” in the same sense that the new C captures $\alpha \times \rho$ —they vary across items at the fixed config and contribute to `sd(item means)`.

The cleaner classification

A component is “captured in the SE” if the estimator’s within-simulation cross-item variance includes it. That is:

anything with an i subscript (items vary).

Blind components are:

- Main effects on factors not in the design (ρ , τ , λ at their chosen fixed levels)
- Interactions between non-item factors at their chosen fixed levels ($\rho\tau$, $\rho\lambda$ at fixed v , h , m)

Under this rule:

Scenario	Captured in SE	Blind
A ($V = 1$, $R = 1$)	α , γ , $\alpha\rho$, $\alpha\tau$, $\alpha\lambda$, ϵ	ρ , $\rho\tau$, $\rho\lambda$
B ($V = 1$, $R = 5$)	α , γ , $\alpha\rho$, $\alpha\tau$, $\alpha\lambda$, ϵ	ρ , $\rho\tau$, $\rho\lambda$
C new ($V = 3$, $R = 5$, item means)	α , γ , $\alpha\rho$, $\alpha\tau$, $\alpha\lambda$, ϵ	$\bar{\rho}$, $\bar{\rho}\tau$, $\bar{\rho}\lambda$ (averaged but blind to variance)
D ($V = 3$, $R = 5$, crossed lmer)	α , γ , $\alpha\rho$, $\alpha\tau$, $\alpha\lambda$, ϵ , $\boldsymbol{\rho}$ (ρ modeled explicitly)	$\rho\tau$, $\rho\lambda$ (still blind at fixed h , m)
E ($V = 3$, $H = 3$, $M = 3$, $R = 5$)	everything	—

Meaningful differences across scenarios

- **B vs C:** same captured set in principle, but C’s “blind” bias is σ_ρ^2/V instead of σ_ρ^2 (smaller by factor V).
- **C vs D:** D explicitly models ρ as a random effect, so its SE includes σ_ρ^2/V as a term.
- **D vs E:** E captures $\rho \times \tau$, $\rho \times \lambda$ via multi- h and multi- m data.